

# Notes on Algebra

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## Contents

|          |                                                                  |           |
|----------|------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Gaussian Elimination</b>                                      | <b>2</b>  |
| <b>2</b> | <b>Matrices &amp; Determinants</b>                               | <b>3</b>  |
| 2.1      | Permutations . . . . .                                           | 5         |
| 2.2      | Rank . . . . .                                                   | 6         |
| 2.3      | The operations of Matrix . . . . .                               | 8         |
| <b>3</b> | <b>Vector Space</b>                                              | <b>10</b> |
| 3.1      | Vector bases and Vector Space Operations . . . . .               | 11        |
| 3.2      | Operators . . . . .                                              | 12        |
| 3.3      | Linear Space Algebra . . . . .                                   | 14        |
| 3.4      | Dual space . . . . .                                             | 16        |
| <b>4</b> | <b>A Short summary from Gaussian elimination to vector space</b> | <b>16</b> |

# 1 Gaussian Elimination

For an augmented matrix that represents a one-to-one correspondence with linear equations.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \longleftrightarrow \begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \cdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}, \quad (1)$$

Assuming the base field  $\mathbb{F}$  so that nonzero elements are invertible, we define the transformations

**Definition 1.1** (*I, II*: Elementary Row Operations).

$$I(i, k) : \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} & b_i \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \mapsto \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ a_{i1} & a_{i2} & \cdots & a_{in} & b_i \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{pmatrix}.$$

$$II(i, k, c) : \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} & b_i \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \mapsto \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1} + ca_{k1} & a_{i2} + ca_{k2} & \cdots & a_{in} + ca_{kn} & b_i + cb_k \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix}.$$

Conducting operations of type *I, II*, the resulting augmented matrix represents a linear system that is equivalent to the original one.

Gaussian elimination can be standardized for an augmented matrix  $A$ ,

$$G(A) : \text{if } a_{11} = 0 \text{ and } \exists i > 1, a_{i1} \neq 0, I(1, i). \\ \forall 1 < i \leq n, II(i, 1, -a_{i1}a_{11}^{-1}).$$

We iterate  $G(A)$ , with the results matrix

$$\bar{A} := \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \cdots & \bar{a}_{1n} & b'_1 \\ 0 & \cdots & \bar{a}_{2i} & \bar{a}_{2(i+1)} & \bar{a}_{2n} & b'_2 \\ 0 & \cdots & \bar{a}_{3k} & \bar{a}_{3(k+1)} & \bar{a}_{3n} & b'_3 \\ \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & \bar{a}_{rs} & \bar{a}_{r(s+1)} & \bar{a}_{rn} & b'_r \\ 0 & 0 & 0 & \cdots & 0 & b'_{r+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & b'_m \end{pmatrix}$$

in which  $1 < i < k < \cdots < s$ .  $\bar{a}_{ij}$  denote the entries after elimination.

**Theorem 1.2.** *The conditions of linear equations ascertaining consistency and determinacy are as follows.*

1. *If rows such as  $(0, 0, \dots, b_i \neq 0)$  do not exist in the result matrix of Gaussian elimination, then the linear equations are consistent.*
2. *If  $r = n$  for consistent linear equations, they are determined.*

*Proof.* We mimic the regression process, observing the result matrix. The main unknown number  $x_{rs}$  is determined by other free unknown numbers  $x_{r(s+1)}, x_{r(s+2)}, \dots, x_{rn}$ . Similarly,  $x_{(r-1)s}$  is determined by  $x_{rs}$  and  $x_{r(s+1)}, x_{r(s+2)}, \dots, x_{rn}$ . Hence, the system is consistent.

If  $r = n$ , then there are no free unknown numbers, and thus, the linear equations are determined, which means

Consistent linear equations are determined  $\iff$  the result matrix in *Row Echelon Form (REF)* satisfies  $r = n$ .

□

From the perspective of mapping, we define

$$T_A : V \rightarrow W, \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \mapsto \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\mathbf{x} \rightarrow A\mathbf{x}}. \quad (2)$$

The result we summarized in Gaussian elimination can be transcribed as:

$$\ker T_A = \ker T_{G(A)} \quad (\text{When } b_i = 0, \text{ the } G(A) \text{ process doesn't alter the solutions of the system}) \quad (3)$$

$$\text{Im } T_A = \text{Im } T_{G(A)} \quad (\text{one-to-one correspondence can be established through the definition of } II) \quad (4)$$

Furthermore, by Gaussian elimination, we can conclude that the matrix for every  $T_A$  is row-equivalent to one *Row Echelon Form*  $\bar{A}$ .

$$\begin{aligned} & \forall T_A : V \mapsto W, \mathbf{x} \longrightarrow A\mathbf{x}, \\ & \iff T_{G(A)} : V \mapsto W, \mathbf{x} \longrightarrow G(A)\mathbf{x}. \end{aligned}$$

If the  $r = n$ , then the matrix for  $T_A$  is row-equivalent to one REF matrix such that there are no  $(0, 0, \dots, b_i)$  rows.

## 2 Matrices & Determinants

**Definition 2.1** (Determinants). *The  $n$ -order determinants are*

$$\det A = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} := \sum_{i=1}^n (-1)^{i+1} a_{i1} M_{i1}.$$

$M_{ij}$ , the minor, is a  $n - 1$ -order determinant that excludes the  $i^{\text{th}}$  row and  $j^{\text{th}}$  column of  $A$ . From the definition,

1.  $\det A$  is a linear function for every row of  $A$

2.  $A \xrightarrow{I(i,k)} A', \det A = -\det A'$

**Theorem 2.2.**

$$A \xrightarrow{I(i,k)} A', \det A = -\det A'$$

$$A \xrightarrow{II(i,k)} A', \det A = \det A'.$$

*Proof.* The proof is followed by the fact that the determinants of matrix  $A$  can be expressed as a linear function of any  $A'$ 's row. It is solvable by induction. □

**Theorem 2.3.** From the last theorem, we discuss the effect of Gaussian elimination on determinants. For matrix  $r = n$ ,

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \xleftrightarrow[I, II]{\text{row-equivalent}} \bar{A} = \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \bar{a}_{1n} \\ 0 & \bar{a}_{22} & \cdots & \bar{a}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \bar{a}_{nn} \end{pmatrix}$$

1.  $\det A = \bar{a}_{11}\bar{a}_{22}\bar{a}_{33}\bar{a}_{44}\cdots\bar{a}_{nn}$ .
2. The linear equations with the coefficient matrix  $A$  are consistent and determined  $\iff \det A \neq 0$ .

**Definition 2.4** (Singular matrix). If a matrix's determinant is 0, then it is singular.

From these attributes of determinants, we are able to deduce the following.

**Theorem 2.5.**  $\forall F : M_{m \times n}(\mathbb{R}) \rightarrow \mathbb{R}, A \mapsto F(A)$  satisfies:

1.  $F(A)$  is a linear function for every row of  $A$ ,
2.  $A \xrightarrow{I(i,k)} A', F(A) = -F(A')$ .

$$F(A) = F(E) \det A.$$

*Proof.* The proof of the theorem is back substitution. We always reduce  $A$  to an upper triangular matrix. We first notice that from  $A \xrightarrow{I(i,k)} A', F(A) = -F(A')$ , we obtain that  $F(A) = (-1)^t F(\bar{A})$  if we use  $I$  process  $t$  times since  $II$  doesn't alter  $F(A)$ . Thus, what's left is

$$F(\bar{A}) = F(E) \bar{a}_{11} \bar{a}_{22} \bar{a}_{33} \cdots \bar{a}_{nn},$$

in which

$$E := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}, \det E = 1.$$

Since  $F(A)$  is a linear function for every row of  $A$ ,

$$\begin{aligned} F(\bar{A}) &= \bar{a}_{nn} \cdot F \left( \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \bar{a}_{1n} \\ 0 & \bar{a}_{22} & \cdots & \bar{a}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \right) \\ &\xrightarrow{\forall 1 \leq i < n, II(i, n, -\bar{a}_{in})} \bar{a}_{nn} \cdot F \left( \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & 0 \\ 0 & \bar{a}_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \right). \end{aligned}$$

The process is a replica of back substitution. Matrix  $\bar{A}$  would eventually be  $E$ . □

## The Column of the determinant

We consider the propagated form of the determinant's definition, which consists of eliminating a random column of the original matrix:

$$F_j(A) = \sum_{i=1}^n (-1)^{i+1} a_{ij} M_{ij}.$$

We can simply conclude that it satisfies Theorem 1.7; thus  $F_j(A) = F_j(E) \det A$ .

$$\begin{aligned} F_j(E) &= \sum_{i=1}^n (-1)^{i+1} a_{ij} M_{ij} = (-1)^{j+1} a_{jj} M_{jj} = (-1)^{j+1} \\ F_j(A) &= (-1)^{j+1} \det A \\ \det A &= \sum_{i=1}^n (-1)^{i+j} M_{ij} \cdot a_{ij} =: \sum_{i=1}^n A_{ij} a_{ij}. \end{aligned}$$

by induction hypothesis:  $M_{jj} = \det E$  for order  $n-1$ ,  $\det E = 1$ .  $A_{ij}$  is the *Algebraic cofactor* of  $a_{ij}$ . The linearity of determinants for columns, with the assistance of algebraic cofactors, implies the viability of Theorem 1.4 on columns.

**Lemma 2.6** (algebraic cofactors acting on linear combinations).

$$\forall k \neq j, \sum_{i=1}^n A_{ik} a_{ij} = 0$$

*Proof.* We construct a matrix  $A'$  such that its  $k^{th}$  column is identical to the  $j^{th}$  column.

Now,  $\det A' = \sum_{i=1}^n A'_{ik} a_{ik} = 0$ , but the  $A'_{ik} = A_{ik}$ ,  $a_{ij} = a_{ik}$ . □

**Theorem 2.7** (Cramer's Rule). *For linear equations with the augmented matrix*

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{pmatrix}, \text{ in which } A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

the  $k^{th}$  unknown number

$$x_k = \frac{\det A_k}{\det A}.$$

The  $A_k$  is constructed by replacing the  $k^{th}$  column of  $A$  to  $(b_1, b_2, \dots, b_n)$ . If  $\det A = 0$ , it is a singular matrix and the linear equations is undetermined or inconsistent.

*Proof.* For  $x_k$ , we may multiply

$$\begin{pmatrix} A_{1k}a_{11} & A_{1k}a_{12} & \cdots & A_{1k}a_{1n} \\ A_{2k}a_{21} & A_{2k}a_{22} & \cdots & A_{2k}a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{nk}a_{n1} & A_{nk}a_{n2} & \cdots & A_{nk}a_{nn} \end{pmatrix}, \begin{pmatrix} A_{1k}b_1 \\ A_{2k}b_2 \\ \vdots \\ A_{nk}b_n \end{pmatrix}.$$

Then, we add every term of the linear equations. Every  $\sum_{i=1}^n A_{ik}a_{ij}$  would cancel out. The only remainder term is  $x_k$ , and its coefficient is  $\sum_{i=1}^n A_{ik}a_{ik} = \det A$ . The right-hand side is  $\sum_{i=1}^n A_{ik}b_i = \det A_k$ . □

## 2.1 Permutations

**Definition 2.8** (even(odd) permutation). *Defining the function  $\varepsilon(\mathbf{J})$ ,  $\mathbf{J} = (i_1, i_2, \dots, i_n)$  is a permutation.*

$$\varepsilon(\mathbf{J}) := \begin{cases} 1 & \text{permutation } \mathbf{J} \text{ is derived by even-numbered transposition from origin} \\ -1 & \text{permutation } \mathbf{J} \text{ is derived by odd-numbered transposition from origin} \end{cases}$$

If  $\varepsilon(\mathbf{J}) = 1$ ,  $\mathbf{J}$  is an even permutation, and vice versa.

**Definition 2.9** (symmetric(antisymmetric) functions). *For a symmetric function  $F(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$ ,*

$$F(\mathbf{a}_1, \dots, \mathbf{a}_i, \dots, \mathbf{a}_j, \dots, \mathbf{a}_n) = F(\mathbf{a}_1, \dots, \mathbf{a}_j, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n).$$

*For an antisymmetric function  $F(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$ ,*

$$F(\mathbf{a}_1, \dots, \mathbf{a}_i, \dots, \mathbf{a}_j, \dots, \mathbf{a}_n) = -F(\mathbf{a}_1, \dots, \mathbf{a}_j, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n).$$

**Theorem 2.10** (A definition of determinant). *The determinant of a  $n$ th-order matrix is a solitary function  $F(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$  of  $n$  rows with  $n$  elements such that*

1. *It is a linear function for each row*
2. *It is antisymmetric*
3.  *$F(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = 1$ , in which  $\mathbf{e}_i$  is the row that has 1 in its  $i$ th position and 0 for its other positions.*

**Theorem 2.11** (multilinear function). *Any multilinear function can be expressed as*

$$F(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n) = \sum_{(i_1, i_2, \dots, i_n)} \alpha_{i_1, i_2, \dots, i_n} a_{1i_1} a_{2i_2} a_{3i_3} \cdots a_{ni_n}$$

in which  $\mathbf{a}_i = (a_{i1}, a_{i2}, a_{i3}, \dots, a_{in})$ , and coefficient  $\alpha_{i_1, i_2, \dots, i_n}$  are only related to their indexes rather than rows. The coefficient  $\alpha_{i_1, i_2, \dots, i_n} = F(\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_n})$ .

(idea) It is easily proven by induction, since any linear function is  $F(\mathbf{x}) = \sum_{i=1}^n a_i x_i$ . In fact, from Theorem 1.13, we acknowledge the explicit formula for determinants. The determinant function

$$F(\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_n})$$

can be considered a permutation  $\mathbf{J}$  of the rows from the original matrix  $E$ . Since we know the  $I$  process adds  $-$  to its determinant, we have  $F(\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_n}) = \varepsilon(\mathbf{J}) \det E = \varepsilon(\mathbf{J})$ . Thus,

$$\det A = \sum_{\mathbf{J}} \varepsilon(\mathbf{J}) a_{1i_1} a_{2i_2} a_{3i_3} \cdots a_{ni_n}.$$

## 2.2 Rank

**Definition 2.12** (Rank of a Matrix). *Let  $A \in M_{n \times n}$ . The rank of  $A$ , denoted by  $\text{rank}(A)$ , is defined as*

$$\text{rank}(A) = r$$

if and only if:

1. *There exists at least one nonzero  $r \times r$  minor of  $A$ ;*
2. *Every  $(r+1) \times (r+1)$  minor of  $A$  is zero.*

Equivalently,  $\text{rank}(A)$  is the largest integer  $r$  for which  $A$  has a nonzero  $r$ -th order minor.

**Theorem 2.13.** *Let  $A \in M_{m \times n}(\mathbb{F})$  and let  $B$  be obtained from  $A$  by the elementary row operation*

$$II(i, j, p).$$

Then

$$\text{rank}(A) = \text{rank}(B).$$

*Proof.* Write

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \xrightarrow{II(i, j, p)} B = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} + pa_{j1} & a_{i2} + pa_{j2} & \cdots & a_{in} + pa_{jn} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Assume

$$\text{rank}(A) = r.$$

Then there exists a  $r \times r$  nonzero minor

$$M = \begin{vmatrix} a_{i_1 j_1} & \cdots & a_{i_1 j_r} \\ \vdots & & \vdots \\ a_{i_r j_1} & \cdots & a_{i_r j_r} \end{vmatrix} \neq 0,$$

where

$$\{i_1, \dots, i_r\} \subset \{1, \dots, m\}, \quad \{j_1, \dots, j_r\} \subset \{1, \dots, n\}.$$

**Case 1: The minor does not involve row  $i$ .**

Then the corresponding minor in  $B$  is unchanged, hence still nonzero. Therefore,

$$\text{rank}(B) \geq r.$$

**Case 2: Every nonzero  $r$ -minor of  $A$  contains row  $i$ .**

Choose such a minor  $M$  and assume  $i = i_k$  for some  $k$ .

Consider the corresponding minor in  $B$ :

$$N = \begin{vmatrix} b_{i_1 j_1} & \cdots & b_{i_1 j_r} \\ \vdots & & \vdots \\ b_{i_r j_1} & \cdots & b_{i_r j_r} \end{vmatrix}.$$

Since rows

$$\mathbf{b}_i = \mathbf{a}_i + p\mathbf{a}_j,$$

and the determinant is linear in each row, we obtain

$$N = M + pM',$$

where  $M'$  is obtained from  $M$  by replacing the  $i$ -th row with the  $j$ -th row:

$$M' = \begin{vmatrix} a_{i_1 j_1} & \cdots & a_{i_1 j_r} \\ \vdots & & \vdots \\ a_{j j_1} & \cdots & a_{j j_r} \\ \vdots & & \vdots \\ a_{i_r j_1} & \cdots & a_{i_r j_r} \end{vmatrix}.$$

Now we analyze  $M'$ . If  $j \in \{i_1, \dots, i_r\}$ , then  $M'$  has two identical rows; hence  $M' = 0$ .

If  $j \notin \{i_1, \dots, i_r\}$ , then the row index set of  $M'$  is

$$\{i_1, \dots, i_{k-1}, j, i_{k+1}, \dots, i_r\},$$

which does not contain  $i$ .

By assumption, every nonzero  $r$ -minor must contain row  $i$ . Therefore, any  $r$ -minor not containing row  $i$  must vanish. Hence  $M' = 0$ .

In both cases,  $N = M \neq 0$ . Thus,  $\text{rank}(B) \geq r$ . Since the row operation is invertible (its inverse is  $II(i, j, -p)$ ), the same argument gives  $\text{rank}(A) \geq \text{rank}(B)$ . Therefore,

$$\text{rank}(A) = \text{rank}(B).$$

□

From now on, we will use index-set minor notation:

$$A[I \mid J] = \begin{pmatrix} a_{i_1 j_1} & a_{i_1 j_2} & \cdots & a_{i_1 j_r} \\ a_{i_2 j_1} & a_{i_2 j_2} & \cdots & a_{i_2 j_r} \\ \vdots & \vdots & & \vdots \\ a_{i_r j_1} & a_{i_r j_2} & \cdots & a_{i_r j_r} \end{pmatrix}, \quad I = \{i_1, \dots, i_r\}, \quad J = \{j_1, \dots, j_r\}.$$

With a little observation, we noticed that the  $\text{rank}(A)$  is exactly the number of non-zero rows in the resulting REF matrix of  $A$ :

$$\bar{A} = \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \cdots & \bar{a}_{1n} & b'_1 \\ 0 & \cdots & \bar{a}_{2i} & \bar{a}_{2(i+1)} & \bar{a}_{2n} & b'_2 \\ 0 & \cdots & \bar{a}_{3k} & \bar{a}_{3(k+1)} & \bar{a}_{3n} & b'_3 \\ \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & \bar{a}_{rs} & \bar{a}_{r(s+1)} & \bar{a}_{rn} & b'_r \\ 0 & 0 & 0 & \cdots & 0 & b'_{r+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & b'_m \end{pmatrix}, \quad \text{rank}(\bar{A}) = \text{rank}(A) = r.$$

From the equation, we can improve Theorem 1.2 to its ranked version:

**Theorem 2.14** (Improve of Theorem 1.2). *Set*

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \text{Augmented Matrix } (A | B) := \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix}$$

The conditions of linear equations ascertaining consistency and determinacy are as follows.

1. If  $\text{rank}(A) = \text{rank}(A | B)$ , then the linear equations are consistent.
2. If  $\text{rank}(A) = n$  for consistent linear equations, they are determined.

The  $\text{rank}(A)$  of matrix  $A$  unfolds the amount of linear information from linear equations with coefficient  $A$ . Rank = number of independent rows.

## 2.3 The operations of Matrix

**Definition 2.15** (Transposition).

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \Rightarrow A^T := \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}, \quad (A^T)_{ij} = a_{ji}.$$

**Definition 2.16** (Multiplication).

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1l} \\ b_{21} & b_{22} & \cdots & b_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nl} \end{pmatrix}, \quad l \leq m.$$

Then,

$$(AB)_{ij} = \sum_{k=1}^m a_{ik} b_{kj} \stackrel{\text{Einstein}}{=} a_{ik} b_{kj}.$$

The multiplication of Matrices resembles Tensor Contraction, which is also conducted by taking the sum of one identical index of two multiplying tensors

$$c_{i_1 i_2 \cdots i_{n-1}} = \sum_{i_k} a_{i_1 i_2 \cdots i_k \cdots i_n} b_{i_1 i_2 \cdots i_k \cdots i_n}.$$



**Theorem 2.17.** *The multiplication of matrices is associative and distributive.*

$$ABC = A(BC), A(B + C) = AB + AC, (A + B)C = AC + BC.$$

$$(AB)^T = B^T A^T.$$

**Lemma 2.18** (Elementary row operation in terms of Matrix Multiplication).

$$I(i, j) \iff T_{ij} := \begin{pmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \cdots & \mathbf{0} & \cdots & \mathbf{1} & \cdots & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & \mathbf{1} & \cdots & \mathbf{0} & \cdots & 0 \\ \vdots & \vdots & & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix}, \quad (T_{ij})_{rs} = \begin{cases} 1 & r = s \neq i, j \\ 1 & (r, s) = (i, j) \text{ or } (j, i) \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

$$II(i, j, c) \iff U_{ij}(c) := \begin{pmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 & \cdots & c & \cdots & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix}, \quad (U_{ij}(c))_{rs} = \begin{cases} 1 & r = s \\ c & (r, s) = (i, j) \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

Put simply, we normalized the manipulations for rows to merely left multiplication of a combination of  $T_{ij}, U_{ij}(c)$ .

**Corollary 2.19.** *By left multiplying a series of  $T_{ij}, U_{ij}(c)$ , we are able to reduce a random matrix  $A$  to its REF. Moreover,*

$$\text{rank}(A) = \text{rank}(T_{ij}A) = \text{rank}(U_{ij}(c)A).$$

**Theorem 2.20.**  $\forall A, B \in M_{n \times n}, \det AB = \det A \det B$ .

*Proof.* Fixed  $B$ , we consider  $F(A) = \det AB$ . The sole issue is to prove its antisymmetry.

$$(AB)_{ij} = \sum a_{ik}b_{kj}. \text{ For row } i, \mathbf{c}_i = \left( \sum a_{ik}b_{k1}, \sum a_{ik}b_{k2}, \sum a_{ik}b_{k3}, \dots, \sum a_{ik}b_{kn} \right).$$

$\implies AB$  is multilinear for rows of  $A$ .

If we switch the  $i, j$  rows of  $A$  ( $I(i, j)$ ), it is equivalent to switch  $i, j$  rows of  $AB$ .

$$\mathbf{c}_i = \left( \sum a_{jk}b_{k1}, \sum a_{jk}b_{k2}, \sum a_{jk}b_{k3}, \dots, \sum a_{jk}b_{kn} \right).$$

$\implies F(A) = -F(T_{ij}A)$ . Antisymmetry is proved.

Thus, we have  $F(A) = F(E) \det A$ , but  $F(E) = \det B$ . □

**Definition 2.21** (Inverse matrix). *For  $A, B \in M_{n \times n}$ , if  $AB = E$ , then  $B =: A^{-1}$  is the inverse matrix of  $A$ .*

*From the definition, we immediately obtain*

1.  $\det A \det B = \det E = 1$ .
2.  $\exists! A^{-1}$ .
3.  $\exists C \in M_{n \times n}, CA = E$ . In fact,  $C = A^{-1}$ .

For the third property of the inverse matrix, we present  $A^T(A^T)^{-1} = E$ . Considering the transposition of the equation  $((A^T)^{-1})^T A = E$ .  $C = ((A^T)^{-1})^T$  exists. Since the association law is valid, uniqueness is obvious.

**Theorem 2.22** (Explicit formula of Inverse Matrix). *We use Cramer's Rule. Considering the vector multiplication of the inverse matrix  $AB = E$ ,*

$$A\mathbf{b}_j = \mathbf{e}_j := (0, 0, \dots, e_j = 1, \dots, 0)^T.$$

*This is a set of linear equations with the same quantity of unknown numbers, so the elements within the inverse matrix is clear:*

$$b_{ij} = \frac{\det A_i}{\det A} = \frac{A_{ji}}{\det A}.$$

*The  $A_i$  is constructed by replacing the  $i^{\text{th}}$  column of  $A$  with  $\mathbf{e}_j$ .*

We summarize the Gauss elimination: a  $n \times n$  non-singular matrix  $A$  can be reduced to REF by left multiplying  $U_{ij}(c), T_{ij}$ . In addition, we notice that

$$\begin{pmatrix} \mathbf{c}_i \\ \mathbf{c}_j \end{pmatrix} \xrightarrow{U_{ij}(1)} \begin{pmatrix} \mathbf{c}_i + \mathbf{c}_j \\ \mathbf{c}_j \end{pmatrix} \xrightarrow{U_{ji}(-1)} \begin{pmatrix} \mathbf{c}_i \\ -\mathbf{c}_i \end{pmatrix} \xrightarrow{U_{ij}(1)} \begin{pmatrix} \mathbf{c}_j \\ -\mathbf{c}_i \end{pmatrix} \xrightarrow{V_j(-1)} \begin{pmatrix} \mathbf{c}_j \\ \mathbf{c}_i \end{pmatrix} \\ \implies T_{ij} = V_j(-1)U_{ij}(1)U_{ji}(-1)U_{ij}(1),$$

where

$$III(i, \alpha) \iff V_i(\alpha) := \begin{pmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & \alpha & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 & 1 \end{pmatrix}, \quad (V)_{ii} = \alpha.$$

It multiplies  $\alpha$  to the  $i$ th row of the matrix  $A$ .

Reducing the square matrix to REF, which is upper triangular form, we left multiply

$$\prod_{j=1}^n \prod_{i \leq j} U_{ij}(-\bar{a}_{ij}\bar{a}_{jj}^{-1}).$$

The product eliminates elements that are not on the diagonal. However, the diagonal matrix

$$\bar{A} = \begin{pmatrix} \bar{a}_{11} & 0 & \cdots & 0 \\ 0 & \bar{a}_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \bar{a}_{nn} \end{pmatrix} = \prod_{i=1}^n V_i(\bar{a}_{ii})E.$$

Thus, we have concluded that:

**Theorem 2.23.**  $\forall A \in M_{n \times n}, \det A \neq 0, A^{-1}$  is a series of products of  $V_i(\alpha), U_{ij}(c)$ , which is the same as Gaussian elimination.

**Corollary 2.24.** Since  $U_{ij}(c), V_i(\alpha)$  are *I, II, III* elementary row operations (They don't alter the rank of the original matrix),  $\forall A \in M_{n \times n}, B \in M_{m \times m}; \det B, \det A \neq 0, \forall C \in M_{n \times m}$ ,

$$\text{rank}(C) = \text{rank}(ACB).$$

### 3 Vector Space

**Definition 3.1** (Vector Space). *The vector space  $L$  is a set that satisfies the following properties:*

1. For  $+$ , a binary operation, the  $(L, +)$  is an Abel group.
2.  $\forall \text{ scalar } \alpha, \alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$ .
3.  $\forall \text{ scalar } \alpha, \beta, (\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}$ .

4.  $\alpha\beta\mathbf{x} = \alpha(\beta\mathbf{x})$
5.  $\forall \mathbf{x} \in L, 1\mathbf{x} = \mathbf{x}$

We therefore present some supplements to the vector space.

**Definition 3.2** (Definitions Involving Subspace).

1.  $\forall L' \subset L, s.t. \forall \mathbf{x}, \mathbf{y} \in L', \alpha\mathbf{x} + \beta\mathbf{y} \in L'. L'$  is the subspace of  $L$ .
2.  $L$  is the sum of its subspaces  $L_1, L_2, \dots, L_k \iff \forall \mathbf{x} \in L, \exists \mathbf{x}_1 \in L_1, \mathbf{x}_2 \in L_2, \dots, \mathbf{x}_k \in L_k,$   
 $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2 + \dots + \mathbf{x}_k.$   
 $L = L_1 + L_2 + \dots + L_k.$
3.  $L$  is the direct sum of its subspaces  $L_1, L_2, \dots, L_k$   
 $\iff L$  is the sum of subspaces  $L_1, L_2, \dots, L_k. \mathbf{x}'$ 's representation in  $\mathbf{x}_1 \in L_1, \mathbf{x}_2 \in L_2, \dots, \mathbf{x}_k \in L_k$  is unique.  
 $L = L_1 \oplus L_2 \oplus \dots \oplus L_k.$

**Definition 3.3** (Dimension of vector space). Similar to the  $\text{rank}(A)$  in matrices, the  $\dim$  is the maximum quantity of linearly independent vectors in vector space  $L$ . If  $d = \dim L$ , then  $\forall \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_{d+1}, \exists \alpha_1, \alpha_2, \dots, \alpha_{d+1}$  such that

$$\begin{aligned} \alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + \alpha_3\mathbf{x}_3 + \dots + \alpha_{d+1}\mathbf{x}_{d+1} &= 0. \\ \alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + \dots + \alpha_{i-1}\mathbf{x}_{i-1} + \alpha_{i+1}\mathbf{x}_{i+1} + \dots + \alpha_{d+1}\mathbf{x}_{d+1} &= \mathbf{x}_i \end{aligned}$$

for at least one  $\alpha_i \neq 0$ . There is one  $\mathbf{x}_i$  that is composed by other vectors. Furthermore,  $\forall n \leq d, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_n$  are, in contrast, linear independent.

**Lemma 3.4.** As in corollary 2.24,

$$\dim(\{A\mathbf{x} : \mathbf{x} \in V\}) = \dim(\{U_{ij}(c)AU_{ij'}^T(c')\mathbf{x} : \mathbf{x} \in V\}).$$

### 3.1 Vector bases and Vector Space Operations

**Definition 3.5** (bases).  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \in L$ , such that they are linear independent, and  $\forall \mathbf{x} \in L, \mathbf{x} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \dots + a_n\mathbf{e}_n$ .

We are able to immediately derive the following fact that

**Theorem 3.6.** For any group of linear independent vectors  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m$ , we can find other  $\mathbf{e}_m, \mathbf{e}_{m+1}, \dots, \mathbf{e}_n$  such that they form a bases of  $L$ .

The result is proved by simply adding linear independent vectors repeatedly, and it can be written in terms of vector spaces  $\forall L' \subset L, \exists L'' \subset L, L = L' \oplus L''$ . The next lemma indicates the dimension's restriction on the vector space.

**Lemma 3.7.** Any linear combinations  $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_m$  of vectors  $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$  such that  $m > n$  are linear dependent.

*Proof.* Since the  $\mathbf{y}_i$  is a linear combination of vectors  $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ , we have

$$\begin{cases} a_{11}\mathbf{x}_1 + a_{12}\mathbf{x}_2 + \dots + a_{1n}\mathbf{x}_n = \mathbf{y}_1 \\ a_{21}\mathbf{x}_1 + a_{22}\mathbf{x}_2 + \dots + a_{2n}\mathbf{x}_n = \mathbf{y}_2 \\ \dots \\ a_{m1}\mathbf{x}_1 + a_{m2}\mathbf{x}_2 + \dots + a_{mn}\mathbf{x}_n = \mathbf{y}_m \end{cases}.$$

We have to construct  $m$  coefficients  $\alpha_1, \alpha_2, \dots, \alpha_m$  that are not all zero and

$$\alpha_1\mathbf{y}_1 + \alpha_2\mathbf{y}_2 + \dots + \alpha_m\mathbf{y}_m = 0. \quad (7)$$

If we plug (8) into the linear combinations, we observe

$$(a_{11}\alpha_1 + a_{21}\alpha_2 + \dots + a_{m1}\alpha_m)\mathbf{x}_1 + (a_{12}\alpha_1 + a_{22}\alpha_2 + \dots + a_{m2}\alpha_m)\mathbf{x}_2 + \dots + (a_{1n}\alpha_1 + a_{2n}\alpha_2 + \dots + a_{mn}\alpha_m)\mathbf{x}_n = 0.$$

The result yields that  $\alpha$ s ensure the validity of the equation (it equals 0), thus we must have

$$\begin{cases} a_{11}\alpha_1 + a_{21}\alpha_2 + \cdots + a_{m1}\alpha_m = 0 \\ a_{12}\alpha_1 + a_{22}\alpha_2 + \cdots + a_{m2}\alpha_m = 0 \\ \cdots \\ a_{1n}\alpha_1 + a_{2n}\alpha_2 + \cdots + a_{mn}\alpha_m = 0 \end{cases}.$$

We solve  $\alpha$ s by Theorem 1.2 since there are fewer equations ( $m$ ) than unknown numbers ( $n$ ).  $\square$

**Corollary 3.8.** *If  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \in L$  are bases, then  $\dim L = n$ .*

Next, we follow the train of thought that the bases is correlated with dimensions.

**Theorem 3.9.**  *$L_1, L_2$  is a vector space.*

$$\dim(L_1 \oplus L_2) = \dim L_1 + \dim L_2,$$

$$\dim(L_1 + L_2) = \dim L_1 + \dim L_2 - \dim(L_1 \cap L_2).$$

*Proof.* Let  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$  and  $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m$  denote a bases of  $L_1, L_2$ . Then, every vector in  $L_1 \oplus L_2$  is

$$\mathbf{x}_1 + \mathbf{x}_2 = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \cdots + a_n\mathbf{e}_n + a_{n+1}\mathbf{f}_1 + a_{n+2}\mathbf{f}_2 + \cdots + a_{n+m}\mathbf{f}_m,$$

$\mathbf{x}_1 \in L_1, \mathbf{x}_2 \in L_2$ .  $\mathbf{x}_1 + \mathbf{x}_2 = 0$  yields the  $a_1, a_2, \dots, a_{n+m} = 0$  since  $\mathbf{x}_1, \mathbf{x}_2 = 0$ , which indicates the linear independence of vectors  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$  and  $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m$ .

The second result can be derived if we set  $L_1 \cap L_2 = L_3, L_1 = L'_1 \oplus L_3, L_2 = L'_2 \oplus L_3$ .  $\square$

**Definition 3.10.** *Let  $V$  be a finite-dimensional vector space over a field  $\mathbb{F}$ .*

*A flag in  $V$  is a sequence of subspaces*

$$\mathbf{0} = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_k \subset V$$

*such that each inclusion is strict.*

*If, in addition,  $\dim V_i = i$  for all  $i = 0, 1, \dots, n$ , where  $n = \dim V$  and*

$$\mathbf{0} = V_0 \subset V_1 \subset \cdots \subset V_n = V,$$

*then the flag is called a complete flag.*

The series of linear spaces is an additive process of new linear independent vectors.

## 3.2 Operators

**Definition 3.11** (Operator). *The linear transformation, or operator, is a map  $A : L \rightarrow M$  such that*

$$A(\mathbf{x} + \mathbf{y}) = A(\mathbf{x}) + A(\mathbf{y}), \quad A(\alpha\mathbf{x}) = \alpha A(\mathbf{x}).$$

The operator is an embodiment of linear algebra in which it not only fully inherits the attributes of matrix algebra, but also surpasses the axis-dependence and is dedicated to the discussion of the algebraic properties of linear spaces.

The operator corresponds to a transformation by a matrix, while a base from  $L, M$  is selected, respectively. Let  $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$  and  $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_m)$  be bases of  $L, M$ , respectively. For any vector in  $L$

$$\begin{aligned} \mathbf{x} &= \alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \cdots + \alpha_n\mathbf{e}_n, \\ A(\mathbf{x}) &= \alpha_1A(\mathbf{e}_1) + \alpha_2A(\mathbf{e}_2) + \cdots + \alpha_nA(\mathbf{e}_n). \end{aligned}$$

If the operator transforms the bases of  $L$  into

$$\begin{cases} A(\mathbf{e}_1) = a_{11}\mathbf{f}_1 + a_{21}\mathbf{f}_2 + \cdots + a_{m1}\mathbf{f}_m \\ A(\mathbf{e}_2) = a_{12}\mathbf{f}_1 + a_{22}\mathbf{f}_2 + \cdots + a_{m2}\mathbf{f}_m \\ \cdots \\ A(\mathbf{e}_n) = a_{1n}\mathbf{f}_1 + a_{2n}\mathbf{f}_2 + \cdots + a_{mn}\mathbf{f}_m \end{cases}, \quad [A]_{\mathcal{F} \leftarrow \mathcal{E}} := \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

then we have if  $A(\mathbf{x}) = \beta_1 \mathbf{f}_1 + \beta_2 \mathbf{f}_2 + \cdots + \beta_m \mathbf{f}_m$ ,

$$\begin{cases} \beta_1 = a_{11}\alpha_1 + a_{12}\alpha_2 + \cdots + a_{1n}\alpha_n \\ \beta_2 = a_{21}\alpha_1 + a_{22}\alpha_2 + \cdots + a_{2n}\alpha_n \\ \cdots \\ \beta_m = a_{m1}\alpha_1 + a_{m2}\alpha_2 + \cdots + a_{mn}\alpha_n \end{cases} \implies [A(\mathbf{x})]_{\mathcal{F}} = [A]_{\mathcal{F} \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}} \quad (8)$$

$[A]_{\mathcal{F} \leftarrow \mathcal{E}}$  is solely determined by the bases  $\mathcal{E}, \mathcal{F}$ , and the operator itself. By converting to matrix  $[A]_{\mathbf{e} \rightarrow \mathbf{f}}$ , we obtain the trivial but important result. Here, we have just introduced a crucial notation  $[\mathbf{x}]_{\mathcal{E}}$ ; it represents the "coordinate" of an unaltered vector  $\mathbf{x}$  under a specific bases  $\mathcal{E}$ . For example, the scalars  $\beta_1, \beta_2, \dots, \beta_m$  in (9) is the coordinates of  $A(\mathbf{x})$  under  $\mathcal{F}$ .

**Lemma 3.12.**  $A : L \rightarrow N, B : N \rightarrow M$ .  $(BA)(\mathbf{x}) := B(A(\mathbf{x}))$ . Let  $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ ,  $\mathcal{G} = (\mathbf{g}_1, \dots, \mathbf{g}_l)$  and  $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_m)$  be bases of  $L, N, M$ , respectively. We reached that

$$(BA)(\mathbf{e}_i) = \sum_{j=1}^m c_{ij} \mathbf{f}_j, \quad c_{ij} = ([B]_{\mathcal{F} \leftarrow \mathcal{G}} [A]_{\mathcal{G} \leftarrow \mathcal{E}})_{ij}.$$

In addition, the  $A(B + C) = AB + AC, (A + B)C = AC + BC$ .

From the fundamental discussion of operators, we instantly resolve the problem involving any shift in coordinates(bases; they are actually equivalent).

**Theorem 3.13** (General Shifts in bases). Let  $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ , and  $\mathcal{E}' = (\mathbf{e}'_1, \dots, \mathbf{e}'_n)$  be two bases of  $L$ . Since the shift in bases does not change the value of a vector, we regard the shift in bases as an Identity Operator  $I_L : L \rightarrow L$ . Then, the matrices

$$[I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}^{-1} = [I_L]_{\mathcal{E} \leftarrow \mathcal{E}'}$$

*Proof.* We recall the expression in (9). Let  $\mathbf{x} \in L$ . If tuple  $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$  transforms to  $\mathcal{E}' = (\mathbf{e}'_1, \mathbf{e}'_2, \dots, \mathbf{e}'_n)$ , then the coordinates alter as

$$[\mathbf{x}]_{\mathcal{E}'} = [I_L(\mathbf{x})]_{\mathcal{E}'} = [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}},$$

which implies that  $[I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}^{-1} = [I_L]_{\mathcal{E} \leftarrow \mathcal{E}'}$ . □

**Theorem 3.14** (Operators' matrix with bases Shifts).  $A : L \rightarrow M$  is an operator. Let  $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ , and  $\mathcal{E}' = (\mathbf{e}'_1, \dots, \mathbf{e}'_n)$  be two bases of  $L$ ;  $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_m)$ , and  $\mathcal{F}' = (\mathbf{f}'_1, \dots, \mathbf{f}'_m)$  be two bases of  $M$ .

$$\begin{array}{ccc} [\mathbf{x}]_{\mathcal{E}'} & \xleftarrow{I_L : L \rightarrow L} & [\mathbf{x}]_{\mathcal{E}} \\ \downarrow A : L \rightarrow M & & \downarrow A : L \rightarrow M \\ [A(\mathbf{x})]_{\mathcal{F}'} & \xleftarrow{I_M : M \rightarrow M} & [A(\mathbf{x})]_{\mathcal{F}} \end{array}$$

From the diagram, we acquire the relation of matrices

$$[A]_{\mathcal{F}' \leftarrow \mathcal{E}'} = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} [A]_{\mathcal{F} \leftarrow \mathcal{E}} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}^{-1}.$$

*Proof.* The main idea is shown in the commutative diagram. In fact, by Lemma 1.39 and Theorem 1.41, we obtain

$$\begin{cases} [A(\mathbf{x})]_{\mathcal{F}'} = [A]_{\mathcal{F}' \leftarrow \mathcal{E}'} [\mathbf{x}]_{\mathcal{E}'} = [A]_{\mathcal{F}' \leftarrow \mathcal{E}'} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}} \\ [A(\mathbf{x})]_{\mathcal{F}} = [A]_{\mathcal{F} \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}} \\ [A(\mathbf{x})]_{\mathcal{F}'} = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} [A(\mathbf{x})]_{\mathcal{F}} \end{cases} \implies [A]_{\mathcal{F}' \leftarrow \mathcal{E}'} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}} = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} [A]_{\mathcal{F} \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}}$$

$$\implies [A]_{\mathcal{F}' \leftarrow \mathcal{E}'} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}} = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} [A]_{\mathcal{F} \leftarrow \mathcal{E}}$$

Because  $I$  is mapping  $L \rightarrow L$ , the matrix  $[I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}$  must be a square matrix with a non-zero determinant. Thus, any  $[I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}$  is nonsingular. □

**Corollary 3.15.** *We note that if  $A$  maps  $L \rightarrow L$ , then the result is reduced to*

$$[A]_{\mathcal{E}' \leftarrow \mathcal{E}'} = [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}'} [A]_{\mathcal{E} \leftarrow \mathcal{E}'} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}'}^{-1}.$$

*The result means that for matrices  $M', M, P$  such that  $M' = PMP^{-1}$ ,  $M, M'$  belong to one operator.*

**Remark 3.16.** *The book usually employs the other perspective of bases shifts, which is, without introducing  $I_L$ , constructing a "shift" operator  $P : L \rightarrow L$  that directly turns the bases.  $P(\mathbf{e}_i) = \mathbf{e}'_i$ . Then,  $[P(\mathbf{x})]_{\mathcal{E}'} = [\mathbf{x}]_{\mathcal{E}}$ .*

**Definition 3.17** (Determinant of an Operator). *For any operator  $A : L \rightarrow L$ , its determinant is defined as  $\det A := \det[A]_{\mathcal{E} \leftarrow \mathcal{E}}$  for any bases.*

### 3.3 Linear Space Algebra

**Definition 3.18** (Isomorphism). *If operator  $A : L \rightarrow M$  is also a bijection (One-to-one correspondence), then it is an isomorphism. We denote  $A : L \xrightarrow{\sim} M$ , and  $L \cong M$ . The Isomorphism is an equivalence relation.*

**Theorem 3.19.** *If linear spaces  $L, M$  have limited dimensions,  $L \cong M \iff \dim L = \dim M$ .*

*Proof.* The necessary condition of the theorem,  $\dim L = \dim M \Rightarrow L \cong M$ , can be proved by using the  $\mathbb{K}^n := \{(a_1, a_2, \dots, a_n) : a_1, a_2, \dots, a_n \in \mathbb{K}\}$ . It clear that  $L \cong \mathbb{K}^n, M \cong \mathbb{K}^n$ , and  $L \cong M$ .

The sufficient condition requires that for a bases  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \in L$ ,  $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n = A(\mathbf{e}_n) \in M$  is also a bases when  $L \cong M$ , which in turn indicates that  $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n$  are linear independent, and  $\forall \mathbf{x} \in M$ ,  $\mathbf{x}$  is a linear combination of  $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n$ . Let  $A \xrightarrow{\sim} M$ .

If  $a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + \dots + a_n \mathbf{e}_n = \mathbf{0}$ , then  $A(a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + \dots + a_n \mathbf{e}_n) = A(\mathbf{0})$ ,  $a_1 \mathbf{f}_1 + a_2 \mathbf{f}_2 + \dots + a_n \mathbf{f}_n = \mathbf{0}' \in M$ , but such a scenario only takes place while  $a_1, a_2, \dots, a_n = 0$  since  $A(\mathbf{0}) = \mathbf{0}'$ . Thus,  $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n$  are linear independent.

For any  $\mathbf{x} \in M$ ,  $\exists! \mathbf{y} \in L, A(\mathbf{y}) = \mathbf{x}$ .  $\mathbf{y} = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + \dots + a_n \mathbf{e}_n$ , so  $\mathbf{x} = A(a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + \dots + a_n \mathbf{e}_n) = a_1 \mathbf{f}_1 + a_2 \mathbf{f}_2 + \dots + a_n \mathbf{f}_n$ . Thus,  $\forall \mathbf{x} \in M$ ,  $\mathbf{x}$  is a linear combination of  $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n$ .  $\square$

**Definition 3.20** (Kernel). *For  $A : L \rightarrow M$ ,  $\ker A := \{\mathbf{x} \in L : A(\mathbf{x}) = \mathbf{0}' \in M\}$ .  $\ker A \subset L$ .*

**Definition 3.21** (Image). *For  $A : L \rightarrow M$ ,  $\text{Im } A := \{A(\mathbf{x}) : \mathbf{x} \in L\}$ .  $\text{Im } A \subset M$ .*

It is time to introduce the Rank-Nullity theorem to reveal the correlation between our previous insights, which we have already seen in the Gaussian elimination and Rank chapters.

**Theorem 3.22** (Rank-Nullity Theorem). *For any linear transformation such that  $T : V \mapsto W$ ,  $\dim V$  is finite,*

$$\dim(\text{Im } T) + \dim(\ker T) = \dim V \quad (9)$$

The creation of the theorem is inspired by (3), (4). We can easily verify some interesting facts: for  $[T(\mathbf{x})]_{\mathcal{E}'} = A[\mathbf{x}]_{\mathcal{E}}$ ,  $\text{rank}(A) = \dim(\text{Im } T)$ ;  $\dim(\ker T)$  is defined as the *nullity* of the mapping  $T$ .

We are able to provide a draft addressing the validity of the Rank-Nullity Theorem, *in my words*.

*Proof.* The linear map  $T$  is depicted by a matrix  $A$  that transforms the vectors in  $V$  to  $W$ ,

$$T : V \rightarrow W, [\mathbf{x}]_{\mathcal{E}} \mapsto [T]_{\mathcal{E}' \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}},$$

$$A := [T]_{\mathcal{E}' \leftarrow \mathcal{E}}, \mathbf{b} := A[\mathbf{x}]_{\mathcal{E}}.$$

If we choose another bases  $\mathcal{F}', \mathcal{E}'$ , then according to Theorem 1.40:

$$A' = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} A [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}^{-1} \implies \text{rank}(A') = \text{rank}(A), \text{ which implies the inalterableness of rank.}$$

$$\mathbf{x} := (x_1, x_2, \dots, x_n)^T \in V, \mathbf{b} := (b_1, b_2, \dots, b_m)^T \in \text{Im } T \subset W, A \in M^{m \times n}.$$

We shall provide some insight for the proof of the theorem one step at a time.

1.  $\dim V = n$ ,  $n$  represents the dimension of the domain  $V$ , which is the number of coordinates of the input vector  $\mathbf{x}$ .
2.  $\dim(\text{Im } T) = \text{rank}(A)$ .

The  $\dim(\text{Im } T)$  represents the number of vectors in a bases in  $\text{Im } T = \{A\mathbf{x} : \mathbf{x} \in V\}$ . We incorporate lemma 1.30 to reduce the matrix  $A$  to Row Echelon Form, and we notice that the result vector of  $\mathbf{b}$  is solely determined by the  $\text{rank}(A)$  main unknown numbers. Thus,  $\dim(\text{Im } T) = \text{rank}(A)$ .

3.  $\dim(\ker T) =$  the quantity of free unknown numbers in the linear equations  $= n - \text{rank}(A)$ .

The  $\ker T \subset V$  is a linear subspace. Observing the Row Echelon Form of  $A$ , we realize that for each  $\mathbf{x}$  such that  $T(\mathbf{x}) = \mathbf{0}$ , its pivot unknown numbers (the first  $\text{rank}(A)$  unknown numbers) are solely determined by its last  $n - \text{rank}(A)$  free unknown numbers according to the linear equations. Thus, its dimension is exactly  $n - \text{rank}(A)$

Therefore, we obtain  $\dim(\text{Im } T) + \dim(\ker T) = \text{rank}(A) + n - \text{rank}(A) = n = \dim V$ .  $\square$

**Remark 3.23.** *The rank-nullity theorem is actually a linear-algebraic version of Isomorphism Theorem A that puts dimension on both sides.*

$$G/\ker \sigma \cong \text{Im } \sigma \implies V/\ker T \cong \text{Im } T, \quad (10)$$

for which

$$\dim V/\ker T = \dim V - \dim(\ker T).$$

My proof is based on rather specific calculations on Matrix; however, the theorem can also be proven in pure operator algebra, refining with extra universality. For such proof to be done, we have to develop our knowledge toward isomorphism.

**Theorem 3.24** (Isomorphic Condition involving Kernel).

$$\forall A : L \rightarrow M, \dim L = \dim M, \ker A = \{\mathbf{0}\} \implies A : L \xrightarrow{\sim} M.$$

Just consider the bases  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \in L$ , and  $A(\mathbf{e}_1), A(\mathbf{e}_2), \dots, A(\mathbf{e}_n) \in M$ .

**Theorem 3.25** (Isomorphic Condition involving Image).

$$\forall A : L \rightarrow M, \dim L = \dim M, \text{Im } A = M \implies A : L \xrightarrow{\sim} M.$$

Consider bases  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \in L$  such that  $A(\mathbf{e}_1), A(\mathbf{e}_2), \dots, A(\mathbf{e}_n) \in M$  forms a bases.

**Theorem 3.26** (Isomorphic Condition involving Singularity). *If for every bases  $\mathcal{E} \in L$ ,  $\mathcal{F} \in M$ ,  $A : L \rightarrow M$  has  $\det([A]_{\mathcal{F} \leftarrow \mathcal{E}}) \neq 0$ , then  $A$  is nonsingular.*

$$A : L \xrightarrow{\sim} M, \dim L = \dim M \iff A \text{ is nonsingular}.$$

**Definition 3.27** (Rank of an operator). *For  $A : L \rightarrow M$ ,  $\text{rank}(A) := \dim(\text{Im } A)$ .*

Now, we have enough prerequisites to provide a general proof of the Rank-Nullity theorem.

**Theorem 3.28** (Extended Rank-Nullity Theorem). *For an operator  $A : L \rightarrow M$ ,  $\text{rank}(A) + \dim(\ker A) = \dim L$ . Plus, we are able to select bases  $\mathcal{E}, \mathcal{F}$  in  $L, M$  such that*

$$[A]_{\mathcal{F} \leftarrow \mathcal{E}} = \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}, \quad r := \text{rank}(A).$$

*Proof.* We decompose  $L = \ker A \oplus L_1$ ,  $M = \text{Im } A \oplus M_1$ . Considering the restriction of  $A$  on  $L_1 \rightarrow \text{Im } A$ , namely  $A' : L_1 \rightarrow \text{Im } A$ ,  $\forall \mathbf{x} \in L, \mathbf{x} = \mathbf{u} + \mathbf{v}$ ,  $\mathbf{u} \in \ker A, \mathbf{v} \in L_1$ ,  $A(\mathbf{x}) = A(\mathbf{u}) + A(\mathbf{v}) = \mathbf{0} + A'(\mathbf{v})$ . Noticing that  $\ker A' = \{\mathbf{0}\}$ ,  $\text{Im } A' = \text{Im } A$ ,

$$\forall \text{ base } \mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_r \in \text{Im } A, \exists \mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_r \in L_1, \mathbf{f}_i = A'(\mathbf{e}_i), \text{ is also a base } \implies A' : L_1 \xrightarrow{\sim} \text{Im } A.$$

We derived  $L_1 \cong \text{Im } A$ , and  $\dim L = \dim(\ker A) + \dim L_1 = \dim(\ker A) + \dim(\text{Im } A)$ .

We select base  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in L_1$ , and  $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{n-r} \in \ker A$ . Combining, we obtain  $\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_{n-r}$  as another base of  $L$ .

Since  $A'$  is an isomorphism,  $A'(\mathbf{v}_1), A'(\mathbf{v}_2), \dots, A'(\mathbf{v}_r) \in \text{Im } A$ , and we just expand these linear independent vectors to a base in  $M$ . Thus, for a base  $\mathcal{F} = A'(\mathbf{v}_1), A'(\mathbf{v}_2), \dots, A'(\mathbf{v}_r), \mathbf{f}_{r+1}, \dots, \mathbf{f}_m \in M$ , we found a base  $\mathcal{E} = \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{n-r} \in L$  such that it satisfied

$$[A]_{\mathcal{F} \leftarrow \mathcal{E}} = \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}.$$

$\square$

**Corollary 3.29.**  $\forall A = M^{m \times n}, \exists$  nonsingular  $B, C$  such that

$$BAC^{-1} = \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}$$

### 3.4 Dual space

## 4 A Short summary from Gaussian elimination to vector space

**Theorem 4.1** (1605, Binet-Cauchy Formula).  $A \in M^{m \times n}, B \in M^{n \times m}$ .

$$C = AB = (c_{ij}), \quad c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad (i, j = 1, \dots, m),$$

$A_{i_1, \dots, i_m}$  selects  $i$  columns of matrix  $A$ ;  $B_{i_1, \dots, i_m}$  extracts  $B$ 's columns.

$$\det(C) = \sum_{1 \leq i_1 < \dots < i_m \leq n} \det A_{i_1, \dots, i_m} \det B_{i_1, \dots, i_m}.$$

If  $m > n$ ,  $\det C = 0$ .

*Proof.* Denote by

$$\mathcal{I} = \{(i_1, \dots, i_m) : 1 \leq i_1 < \dots < i_m \leq n\}.$$

#### 1. Leibniz expansion of $\det C$ .

Using Leibniz formula,

$$\begin{aligned} \det C &= \sum_{\sigma \in S_m} \varepsilon(\sigma) \prod_{l=1}^m c_{l\sigma(l)} = \sum_{\sigma \in S_m} \varepsilon(\sigma) \prod_{l=1}^m \sum_{k=1}^n a_{lk} b_{k\sigma(l)}. \\ \det C &= \sum_{\sigma \in S_m} \varepsilon(\sigma) \sum_{1 \leq k_1, \dots, k_m \leq n} \prod_{l=1}^m a_{lk_l} b_{k_l \sigma(l)}, \end{aligned}$$

by expanding the product.

#### 2. Cancellation of terms with repeated indices.

If  $(k_1, \dots, k_m)$  contains repeated values, say  $k_p = k_q$  with  $p \neq q$ , then exchanging  $p$  and  $q$  does not change the product

$$\prod_{l=1}^m a_{lk_l} b_{k_l \sigma(l)},$$

but changes the sign of the permutation  $\sigma$ . Hence such terms cancel in pairs.

Therefore, only tuples  $(k_1, \dots, k_m)$  with all  $k_l$  distinct contribute.

#### 3. Reindexing by subsets and permutations.

If  $k_1, \dots, k_m$  are all distinct, then they form a subset  $I = \{i_1, \dots, i_m\} \in \mathcal{I}$ , and there exists a unique permutation  $\tau \in S_m$  such that  $k_l = i_{\tau(l)}$ . Thus, we may rewrite

$$\begin{aligned} \sum_{k_1, \dots, k_m \text{ distinct}} &\iff \sum_{I \in \mathcal{I}} \sum_{\tau \in S_m} \\ \det C &= \sum_{\sigma \in S_m} \varepsilon(\sigma) \sum_{I \in \mathcal{I}} \sum_{\tau \in S_m} \prod_{l=1}^m a_{li_{\tau(l)}} b_{i_{\tau(l)} \sigma(l)}. \end{aligned}$$

#### 4. Rearranging permutations.

Let  $\rho := \sigma \circ \tau^{-1}$ . Then  $\sigma = \rho \circ \tau$ , and  $\varepsilon(\sigma) = \varepsilon(\rho)\varepsilon(\tau)$ . Hence

$$\det C = \sum_{I \in \mathcal{I}} \sum_{\rho \in S_m} \varepsilon(\rho) \left( \sum_{\tau \in S_m} \varepsilon(\tau) \prod_{l=1}^m a_{li_{\tau(l)}} \right) \left( \prod_{k=1}^m b_{i_k \rho(k)} \right).$$



## 5. Identification with determinants.

Observe that

$$\sum_{\tau \in S_m} \varepsilon(\tau) \prod_{l=1}^m a_{l \ i_{\tau(l)}} = \det(A_{i_1, \dots, i_m}),$$

$$\sum_{\rho \in S_m} \varepsilon(\rho) \prod_{k=1}^m b_{i_k \ \rho(k)} = \det(B_{i_1, \dots, i_m}).$$

Therefore,

$$\det C = \sum_{I \in \mathcal{I}} \det(A_{i_1, \dots, i_m}) \det(B_{i_1, \dots, i_m}).$$

If  $m > n$ , there are no such subsets  $I$ , hence  $\det C = 0$ . □